Xiaoyan "Alex" Guo

alex.xy.guo@gmail.com | 858.228.7136

Education

2024 – present	Ph.D. in Computational Biology & Medicine **(Joint program of Weill Cornell Medicine, Memorial Sloan Kettering Cancer Center, and Rockefeller University)**	**Tri-Institutional PhD Program**
2020 – 2022	M.S. in Computational Biology	**Carnegie Mellon University**
2016 – 2020	B.S. in Bioengineering: Bioinformatics	**University of California, San Diego**

Professional Experiences

Ph.D. Candidate @ Weill Cornell Medicine May 2026–Present

Reznik Lab, Memorial Sloan Kettering Cancer Center – New York, NY

- Analyzing large-scale clinical, genomic, body-composition imaging, and paired human tissue metabolomics data using statistical modeling, machine learning, and survival analysis
- Conducting computational oncology research to identify physiological and genomic determinants of cancer therapy-related adverse events, and to define metabolic programs associated with clear cell renal cell carcinoma metastasis
- Investigating associations between patient characteristics and treatment toxicity, quantifying metabolic relationships among primary tumors, metastatic tumors, and host tissues, and developing models of metastatic progression

Associate Computational Biologist II @ Broad Institute Jun 2022–May 2024

Mootha Lab, Broad Institute of MIT & Harvard – Cambridge, MA

- Performed integrative analysis of data from various technologies (genomics, transcriptomics, proteomics, CRISPR screens, metabolomics, microscopy images, etc.) to find biomarkers and pathways related to mitochondrial diseases and biology
- Designed computational methods for predicting mitochondrial protein interactions and investigating biological mechanisms of under-studied parasite organisms using proteomics and transcriptomics data for HHMI's Emerging Pathogens Initiative
- Created unsupervised methods for quantifying regional neuron density of mouse brain images
- Developed a bioinformatics pipeline to analyze results from a novel single-cell combinatorial-indexing mitochondrial genome sequencing technology developed in-house

Data Scientist - Computational Biology Summer Intern @ Helomics May 2021–Aug 2021

Pittsburgh, PA

- Built an image processing pipeline with scikit-image and OpenCV for unsupervised analysis of Hematoxylin & Eosin stained whole slide images of tumor tissues
- Conducted thorough literature reviews on histopathologic image analysis and implemented state-of-the-art methods for extracting clinically meaningful features from images
- Built a random forest model for predicting drug response of ovarian cancer patients using features extracted from pathology images

Innovation Center Technical Intern @ BIOVIA, Dassault Systemes Jun 2019–Sep 2019

La Jolla, CA

- Implemented an unsupervised pipeline to extract biochemically meaningful features from biological sequence data
- Developed an algorithm for the fast homology-based search for similar sequences (DNA, RNA, amino acid)
- Trained XGBoost and Random Forest models for enzyme activity prediction using biochemical features extracted from amino acid sequences

Publications

[1] Z. Bakouny*, **X. A. Guo***, F. Huang*, S. Mohan*, R. Walser*, Z. Lu, T. Perea-Chamblee, L. Khan, N. Ahmed, A. Ocejo, A. A. Hakimi, N. Shah, M. H. Voss, K. C. Arbour, Y.-M. M. Cheung, H. Azhari, D. M. Faleck, R. Niec, M. T. A. Donoghue, J. J. Orgera, A. Syed, M. F. Berger, M. Waters, K. Pichotta, C. Fong, J. Jee, N. Schultz, D. Schrag, L. Yarmus, A. J. Schoenfeld, A. Gusev, R. R. Kotecha, R. J. Motzer, C. B. Thompson, W. Tansey, J. Carrot-Zhang, and E. Reznik. "*Population-scale precision safety in oncology reveals clinical and genetic determinants of systemic therapy toxicity*". In: ***In major revision at Nature****; preprint at medRxiv, doi:10.1101/2026.09.16.26363259* (2026).

[2] S. Boscenco, V. J. Castillon, J. Wang, E. Tse, S. Freeman, Z. Bakouny, S. Mohan, **X. A. Guo**, R. Walser, J. Song, C. P. Zambirinis, L. Bojmar, R. R. Kotecha, M. Hilmi, M. S. May, G. A. Vitiello, T. Janowitz, M. D. Goncalves, N. Gangai, D. Lyden, A. Z. Herskovits, P. Iyengar, W. R. Jarnagin, R. E. Schwartz, R. Sosa, J. Jee, E. M. O'Reilly, N. Schultz, S. P. Shah, W. Park, J. W. Garrett, P. J. Pickhardt, N. C. Swinburne, and E. Reznik. "*Anatomical dynamics define cancer cachexia subtypes and identify systemic inflammation as a marker of lethal wasting*". In: ***Accepted at Cell*** (2026).

[3] J. A. Stefely, F. G. Deng, M. Z. Chen, K. Nguyen, B. E. Luce, M. Y. Fry, **X. A. Guo**, K. Maliaris, K. D. Rivera, M. Papanastasiou, T. M. Issa, B. Kanakapura Sundararaj, S. L. Y. Root, D. R. Mani, R. Aphasizhev, I. Aphasizheva, C. N. Betsinger, S. A. Carr, M. T. Duraisingh, M. Goyal, M. Kim, C. G. Lian, Y.-T. Liao, S. Y. Mian, A. S. Paul, M. Solayman, A. F. Vacas, L. H. Chao, S. E. Calvo, N. D. Udeshi, J. Samuelson, and V. K. Mootha. "*Mitochondrial proteome of Acanthamoeba delineates aerobic and anaerobic pathways dynamically regulated by oxygen*". In: ***Cell*** 189.20 (Oct. 2026), 6246–6266.e19.

[4] M. Miranda, T. J. Durham, C. R. Vanderburg, N. M. Nadaaf, **X. A. Guo**, O. Goldberger, P. Lichtenegger, M. A. Walker, O. S. Skinner, E. Macosko, and V. K. Mootha. "*Single-cell analysis of Leigh syndrome brain lesion formation and its reversal by hypoxia*". In: ***In review at Neuron*** (2026).

[5] A. Kotrys, T. Durham, **X. A. Guo**, V. R. Vantaku, S. Parangi, and V. K. Mootha. "*Single cell analysis reveals context-dependent, cell-level selection of mtDNA*". In: ***Nature*** 629.8011 (May 2024), pp. 458–466.

[6] R. S. Rogers, R. Sharma, H. Shah, O. Skinner, **X. A. Guo**, A. Panda, R. Gupta, T. J. Durham, K. B. Shaughnessy, J. R. Mayers, K. A. Hibbert, R. A. Baron, B. T. Thompson, and V. K. Mootha. "*Circulating N-lactoyl-amino acids and N-formyl-methionine reflect mitochondrial dysfunction and predict mortality in septic shock*". In: ***Metabolomics*** 20.2 (Mar. 2024), p. 36.

[7] P. R. Joshi, S. Sadre, **X. A. Guo**, J. G. McCoy, and V. K. Mootha. "*Lipoylation is dependent on the ferredoxin FDX1 and dispensable under hypoxia in human cells*". In: ***Journal of Biological Chemistry*** 299.9 (July 2023), p. 105075.

[8] H. Lei*, **X. A. Guo***, Y. Tao, K. Ding, X. Fu, S. Oesterreich, A. V. Lee, and R. Schwartz. "*Semi-deconvolution of bulk and single-cell RNA-seq data with application to metastatic progression in breast cancer*". In: ***Bioinformatics*** 38.Suppl 1 (June 2022), pp. i386–i394.

[9] J. Vu, E. Cisneros, H.Y. Lee, L. Le, Q. Chen, **X. A. Guo**, R. Rouzbehani, J. Jankovic, S. Factor, C. G. Goetz, R. L. Barbano, J. S. Perlmutter, H. A. Jinnah, S. Pirio Richardson, G. T. Stebbins, R. Elble, C. L. Comella, and D. A. Peterson. "*Head tremor in cervical dystonia: Quantifying severity with computer vision*". In: ***Journal of the Neurological Sciences*** 434.120154 (Mar. 2022), p. 120154.

(* co-first authors)

Research Experiences

Tri-Institutional Joint PhD Program in Computational Biology & Medicine — Aug 2024–Present

Identifying physiological and genomic risk factors of immune-related adverse events

[Mentor: Dr. Ed Reznik @ Memorial Sloan Kettering Cancer Center]

- Defining physiological and genomic features associated with elevated risk of immune-related adverse events
- Uncovered associations between human leukocyte antigen (HLA) alleles and specific adverse events such as adrenal insufficiency and liver toxicity
- Identified a link between patient body composition and increased risk of cancer therapy-related hypothyroidism
- Co-first author on manuscript in major revision at *Nature*; preprint: doi.org/10.1101/2026.09.16.26363259

Defining metabolic programs of clear cell renal cell carcinoma metastasis using paired human tissue metabolomics

[Mentor: Dr. Ed Reznik @ Memorial Sloan Kettering Cancer Center]

- Defining tumor-intrinsic metabolic reprogramming associated with distant metastasis
- Quantifying metabolic coupling between metastatic tumors and adjacent normal tissues at the host site
- Constructing a primary-to-metastasis metabolic trajectory to test whether a tumor's position on the axis is associated with progression risk and shorter time-to-metastasis

Rotation 3: Profiling neuroendocrine–non-neuroendocrine plasticity in small cell lung cancer with single-cell multiomics data

[Mentor: Dr. Joe Chan @ Memorial Sloan Kettering Cancer Center]

- Analyzed 10x multiome (scRNA-seq/scATAC-seq) data from 11 patient-derived xenograft samples to characterize NE to/from non-NE state transitions
- Inferred pseudotime trajectories in both directions (starting from NE vs SCLC-N/NonNE) and evaluated whether gene-expression dynamics were consistent with expected biology
- Integrated RNA and chromatin data along pseudotime by binning cells and quantifying ASCL1/NEUROD1 target gene expression and target-region accessibility using SCENIC+ regulatory network inference

Rotation 2: Identifying nuclear-mitochondrial tumor genome co-mutation patterns in health-system-scale clinical sequencing dataset

[Mentor: Dr. Ed Reznik @ Memorial Sloan Kettering Cancer Center]

- Recycled off-target reads from large public targeted clinical sequencing cohorts to call mtDNA somatic mutations
- Identified an accumulation of silent mtDNA mutations in *NPM1*-mutated AML

Rotation 1: Profiling structure-specific transcriptomic changes in lymph nodes with cellular resolution spatial transcriptomics data

[Mentor: Dr. Jan Krumsiek @ Weill Cornell Medicine]

- Conducted QC and analysis of NanoString CosMx single-cell spatial RNA and protein expression data
- Applied a variational autoencoder with spatial prior to identify histological structure in organ biopsies, such as germinal centers
- Identified transcriptomic changes associated with developing B cells located in different spatial niches

Broad Institute of MIT & Harvard — Jun 2022–May 2024

Integrative analysis of complexome profiling and time-course RNA-seq reveals mitochondrial dynamics of under-studied eukaryotic parasites

[Mentor: Prof. Vamsi Mootha]

- Developed computational methods for the semi-supervised prediction of mitochondrial protein interactions from complexome profiling (BN-PAGE-MS) data
- Processed large proteomics datasets from the Protein Data Bank to create a training dataset
- Analyzed paired time-course RNA-seq and TMT proteomics data to identify distinct stages of mitochondrial activity
- Resulting paper published in *Cell* (2026)

Large-scale, single cell analysis of mtDNA heteroplasmy reveals dynamics of environment-dependent selection

[Mentor: Prof. Vamsi Mootha]

- Developed a bioinformatics pipeline for processing combinatorially indexed Illumina FASTQ reads into a human-readable single-cell mtDNA heteroplasmy spreadsheet suitable for downstream analysis

Identifying metabolomic predictors of mitochondrial dysfunction and mortality in septic shock

[Mentor: Prof. Vamsi Mootha]

- Conducted metabolomics data analysis and QC, based on cohort demographics and clinical metrics
- Built linear models adjusted for sex and age to identify significantly changing metabolites in septic shock patients compared to ambulatory controls
- Identified 9 candidate metabolic biomarkers significantly elevated in septic shock

Quantifying regional neuronal density in mouse brains affected by Leigh syndrome

[**Mentor: Prof. Vamsi Mootha**]

- Developed unsupervised methods for neuron counting using immunofluorescence microscopy images of mouse brains
- Registered whole-brain images to the Allen Brain Atlas to enable mouse-to-mouse comparison of regional neuronal densities

Systematic profiling of the mitochondrial proteome in HepG2 cells with iron-sulfur cluster disruptions in hypoxic conditions

[**Mentor: Prof. Vamsi Mootha**]

- Conducted TMT proteomics data analysis, visualization and QC
- Performed statistical tests to find proteins and mitochondrial pathways significantly affected by iron-sulfur cluster disruptions and/or hypoxic conditions
- Link to paper: bit.ly/fdx1-paper

Carnegie Mellon University — Apr 2021 – Jun 2022

Automated analysis of expansion microscopy tumor tissue images for cancer pathology

[**M.S. Thesis, Mentors: Prof. Russell Schwartz & Prof. Yongxin Leon Zhao**]

- Developed a Python toolkit for unsupervised nuclei segmentation and human-interpretable morphological feature extraction from fluorescent expansion microscopy images
- Implemented computational methods to quantify RNA expression from multiplexed fluorescence in situ hybridization images
- Designed a weakly-supervised deep learning framework for identifying informative sub-regions of tumor tissue microscopy images for predicting drug responses in muscle-invasive bladder cancer
- Link to thesis: bit.ly/thesis-ms-aguo

Semi-deconvolution of bulk and single-cell RNA-seq data with application to metastatic progression in breast cancer

[**Mentor: Prof. Russell Schwartz**]

- Developed an objective function that uses scRNA-seq data as a reference for solving the bulk RNA-seq deconvolution problem
- Studied the relationship between average gene expression and gene drop-out rate to correct scRNA-seq zero-inflation during multi-modal data integration
- Applied a deconvolution algorithm on breast cancer metastasis dataset to identify changes in cell-type-specific gene expression and cell-type abundances
- Link to paper: bit.ly/rads-paper

University of California, San Diego **Jul 2018 – Jun 2020**

Machine learning classification for diagnostics of leukemia based on single cell flow-cytometry data analysis

[B.S. Senior Design, Mentor: Prof. Yu Max Qian]

- Developed a PyTorch-based pipeline for weakly-supervised identification of diagnostically informative cell populations from single-cell flow cytometry data
- Implemented gradient descent algorithms with regularization to optimize classification of leukemia vs healthy cell populations

Evaluating range of motion in cervical dystonia with computer vision

[Mentor: Dr. David Peterson]

- Designed a novel scoring system for evaluating the range of motion of dystonia patients using features extracted with a facial landmark detection algorithm
- Built an Electron.JS desktop software application to automatically assess the severity of cervical dystonia in patients
- Link to slides: bit.ly/rom-sejnowski2020

Presentations

- **"Host organ environment shapes clear cell renal cell carcinoma metastatic metabolism"** (Poster) 2026 American Association for Cancer Research Annual Meeting. April 22, 2026.
- **"Hunting for Nuclear-Mitochondrial Co-Mutation Patterns in MSK-IMPACT"** (Poster) 2025 Halvorsen Center for Computational Oncology Retreat. June 20, 2025.
- **"A novel pipeline of BN-PAGE-MS data analysis for identifying mitochondrial protein-protein interactions in organisms with under-studied proteome"** (Poster) HHMI Emerging Pathogens Initiative Mito-EPI Symposium 4. April 5, 2024.
- **"Systematic analysis of a protozoan encystation time-course RNA-seq dataset"** (Oral) HHMI Emerging Pathogens Initiative Mito-EPI Symposium 3. November 3, 2023.
- **"Development of a pipeline for mitochondrial complexome analysis by BN-PAGE-MS"** (Oral) HHMI Emerging Pathogens Initiative Mito-EPI Symposium 2. July 7, 2023.
- **"Semi-deconvolution of bulk and single-cell RNA-seq data with application to metastatic progression in breast cancer"** (Oral) The Annual International Conference on Intelligent Systems for Molecular Biology (ISMB). July 12, 2022. Link to slides: bit.ly/rads-ismb2022

Awards and Honors

- **Travel Fellowship** ($750), Intelligent Systems for Molecular Biology (ISMB), 2022
- **Research Honors & Excellence** from Carnegie Mellon University, 2022
- **Provost Honors** from University of California, San Diego, 2019/2020

Skills

Data Science	PyTorch, NumPy, SciPy, scikit-learn, scikit-image, OpenCV, pandas, lifelines
Bioinformatics	Scanpy, SCENIC+, HIBAG, featureCounts, Salmon, Cell Ranger, edgeR, DESeq2
Programming	Python, R, Bash, MATLAB, Java, JavaScript, C, C++, Git

Teaching Assistant

- **02-750 Automation of Scientific Research (on the topic of Active Learning)** @ Carnegie Mellon University, Spring 2022

Misc.

Student Mentor @ UCSD Biomedical Engineering Society 2019 - 2020

- Shared academic and professional experiences with younger students to help their professional development

Outreach Volunteer @ Society of Women Engineers - San Diego 2017

- Mentored high school girls on potential careers in science and technology
- Taught introductory Python and HTML/CSS to beginners in programming
- Demonstrated the application of bioengineering through the construction of a prosthetic hand